NYC UNDER THE SPOTLIGHT

EXPLORING THE 2021 NYPD ARREST DATASET

Introduction

This document presents a data-driven analysis of the 2021 NYPD arrest dataset, conducted by a team of five data analysts to uncover key trends in enforcement activity across New York City. After cleaning and validating our data, our analysis explores how arrests vary by borough, age group, gender, race, and offense type. Through data visualization created using Tableau and exploratory analysis, the team identified geographic concentrations of arrest activity, highlighted disparities across demographic groups, and examined patterns in high-frequency offenses. These findings translate complex data into actionable insights, providing a foundation for more informed and equitable decision-making in public safety strategy and resource allocation.

Methodology

Data Source

Chose dataset from COOP, originally in Excel format.

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Data Cleaning

- Removed incomplete Records
- Handled Missing Offense Levels
- Verified data integrity
- Preserved ambiguous columns

Exported to an Excel Sheet

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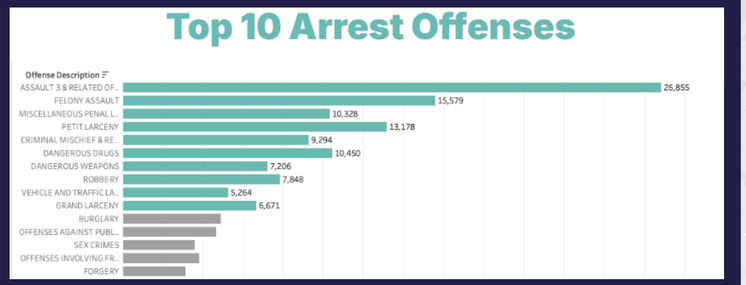
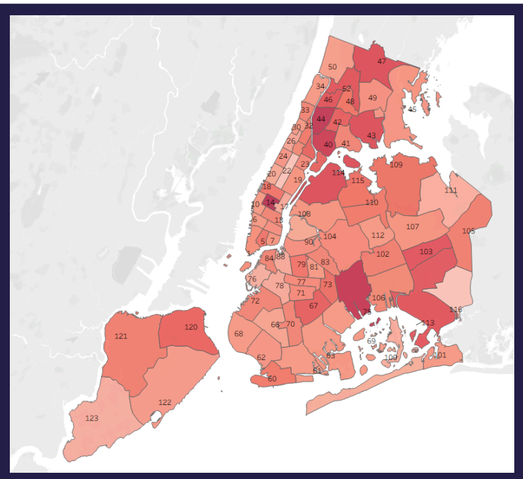
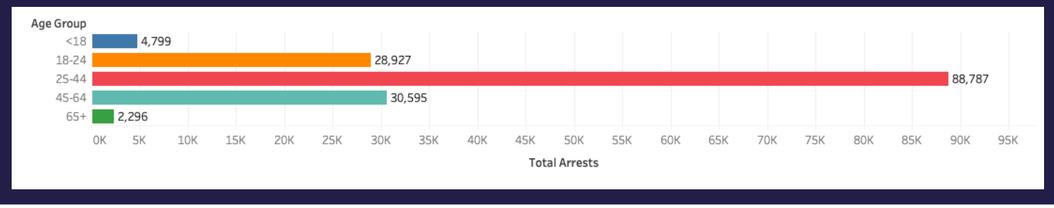
Exploratory Analysis

Pull Descriptive Stats

Created Visualizations on Tableau

Findings

- Arrest activity is highest in Brooklyn, followed by Manhattan and the Bronx, indicating strong geographic concentration.
- Arrest "hotspots" are clustered within specific precincts, with precincts 14, 44, and 75 showing the highest activity
- Adults aged 25–44 represent the largest share of arrests, making them the primary demographic group
- Male individuals account for the vast majority of arrests (over 80%), far exceeding female arrests
- Black individuals make up nearly half of all arrests, followed by Hispanic and White populations, showing notable disparities across racial groups, suggesting differences in enforcement patterns
- Arrests are heavily concentrated in a few offense categories, particularly:
  - Assault-related offenses (most frequent)
  - Theft and larceny crimes
  - Drug-related offenses

Conclusions & Recommendations

Our analysis shows that arrest patterns across New York City vary significantly by borough, demographic group, age group, and offense type, with activity heavily concentrated in areas such as Brooklyn, Manhattan, and the Bronx. These findings also highlight notable demographic disparities, suggesting differences in enforcement patterns, while reinforcing that arrest data reflect policing activity rather than actual crime rates alone. Based on these insights, resources such as lighting, emergency infrastructure, and patrol efforts should be strategically focused in high-activity areas. Additionally, combining data with community context is essential to reduce potential bias and ensure fair enforcement. Prioritizing prevention efforts for the most impacted groups, along with continued improvements in transparency and data monitoring, can support more targeted and equitable public safety decisions across the city.